RESEARCH DATA

A Revolutionary Chlorophyll Enhancer Product Based on MS III n Formula
Developed By Dr RD Sheth At Gujarat Agriculture University

BIO MAGIC

DRISHTI BIOTECH

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The Research data indicate the superiority of BIO MAGIC -3 sprays at intervals of 15 days commencing from 30 DAS registering 31% increase in yield. This experiment also showed that conventional treatments of FeSo4+ Citric Acid spray was not as effective in increasing yield as BIO MAGIC , The data also occur with earlier recommendation adopted at Joint Agresco, May 1993, when a 28% increase in yield was recorded. Considering the results ,i.e. three years in Agronomy Department (from where the recommendation emerged), two years collecting data from experiments undertaken based on suggestion from committee on chlorosis in groundnut regulators, BIO MAGIC has proved itself an effective agent in increasing the yield of groundnut by between 25 to 30%.

BIO MAGIC
An organic based bio –stimulant with plant extracts- Treats chlorosis, increases growth and yield

REPORT ON REASEARCH, EXPERIMENTES & TESTS CONDUCTED

INTRODUCTION.

Chlorosis is a common occurrence all over the world (40% of world soils- Chen & Bank 1982, Shorrocks 1984). Groundnut is considered one of the most sensitive crops which suffer heavily from his bight sometimes resulting in 100% damage. The most pronounced effect is a drastic reduction in chlorophyll very much essential for biomass production (Lindsay, 1982, 1984). It has been reported that at all the stages, i.e., flower production, peg initiation, peg penetration, pod initiation, pod formation, an pod filling adequate supply of photosynthate is essential (Williams et al 1975, 1978 and Krishnashastri et al 1985). At the international level 60 institutions are engaged in researching cost effective solutions to this problem and even after decades of research iron deficiency remains an important plant nutrition problem (Chaney & Bell 197, Katyal & Sharma 1985).

MANAGEMENT

Two principal methods of treating iron deficiency are followed:

A. Soil Treatment : In soil treatment with iron containing fertilizers synthetic iron chelates are effective but because of the very high cost (in India Rs 2.5 Lakhs/ha) they are cost prohibitive. Application of ferrous sulphate does not give repeat performance and also large quantities are required.

B. Foliar Treatment : Foliar sprays of inorganic salts have been found to be of benefit but require frequent treatment. In the category of foliar spraying of 2 % ferrous sulphate and ferrous sulphate + Citric acid (5:1 of 10:1) are recommended but farmers have not fully adopted this method as on many occasions phyototoxic effect with stunting of plants have been observed.

Most frequently it is observed that in soils with iron deficiency, zinc and manganese deficiencies have also been noted to occur. BIO MAGIC takes care of these deficiencies have also been noted to occur. BIO MAGIC takes care of simultaneously suffer from nitrogen deficiency as nitrogen fixing enzymes also require iron. In Cyprus a 45% increase in yield of groundnut was obtained by foliar spray (Papastylianou 1990). In Thailand a 60% increase was recorded by foliar spray (Ratanrat et al 1988). There are several other reports where significant results have been achieved with appropriate foliar sprays. Dangule & Zende (1976) reported that foliar application of Miscnelf (Ms 10) in two sprays at 35 and 50 DAS increased groundnut yield by about 12% over the control. After considering all the above aspects, the manufactures tried to develop sequestered compounds having balanced proportion of micronutrients appropriately blended with microbial extract and chelating agent. Results of BIO MAGIC were very encouraging in pto and mini –plot experiments as well as in 22 field trials conducted in problem areas. Results indicated a 28% increase in yield in all there moisture regimes tired. According to current available literature on iron chlorosis, the general cocenus amongst scientists is that foliar spray of iron containing formulations is the most practical remedy having both the advantage of cost – effectiveness and avoiding the need for frequent applications. BIO MAGIC is the answer. It combines the advantages of being cheap, very concentrated and requiring few applications.

EFFECT OF BIO MAGIC ON PADDY AND CHILLY CROPS

Seeing the very good effect of BIO MAGIC in correcting chlorosis and improving health and final yield of groundnut, it was decided to test its efficacy on paddy and chilies. Paddy seedlings and Paddy showing symptoms of chlorosis totally recovered after just now spray applications of BIO MAGIC is the leaves turned dark green and the whole plant appeared healthier. The same results were obtained in trails with chillies.

REASEARCH DATA

Tabulated hereunder are the results of some of the data culled from field trails. It will be noted that every effort has been made to compare BIO MAGIC with other treatments and also the different types of application of BIO MAGIC as well as the application of BIO MAGIC in various concentrations TO determine the best concentration and method of application.

Table 1
RESULTS OF BIO MAGIC ON VARIOUS DEMONSTRATIONS ON FARMERS' FIELDS IN INDIA

DISTRICT	AGENCY	NUMBEROF DEMONSTRATIONS	INCREASE IN YIELD
Junagadh	Aga Khan Rural Support prog. Training & Welfare Scheme	29	18%
Bhavnagar	Government of Gujarat Training & Welfare Scheme	8	12%
Rajkot	Government of Gujarat Training & Welfare Scheme	25	12%
Kheda	Government of Gujarat Training & Welfare Scheme	14	15%

In fields where there was proper rainfall and adequate sunshine the increase in yield was above 25% .In on field in the Kheda districts there was an increase of 5o% in yield and Rajkot one field showed a 27% yield increase despite there being no rainfall.

Table 2
EFFECT ON BIO MAGIC ON VARIOUS PARAMETERS OF GROUNDNUT CULTIVATION

RESULTS	WITH BIO MAGIC (TRIAL)	WITHOUT BIO MAGIC (CONTROL)	CHANGE (% Age)
Number of roots per plant	36.60	28.15	+30.02%
Number of pods per plant	29.80	12.51	+138.21%
Average Weight of each pod	10.86	8.96	+21.21%
Weight of pods per Ha (kg)	3104	2591	+19.80%
Percentage of kernel	72.34	66.35	+9.03%
Percentage of oil	49.86	48.56	+2.88%
Percentage of protein	24.89	20.41	+21.95%
Percentage of carbohydrate	30.37	25.08	+21.09%
Production per hectare (kg)	1114.00	860.00	+29.53%
TOTAL	102.60	66.60	+54.05%

TABLE -3
CHLOROPHYLL CONTENT IN SAPOTA LEAVES TREATED WITH BIO MAGIC (Sapota Orchard Test Farm , Anand , Gujarat , India,1995)

Sample	**CONTROL GROUP**			**TEST GROUP**			**Change**
	Chlorophyll A	Chlorophyll B	Total Chlorophyll	Chlorophyll A	Chlorophyll B	Total Chlorophyll	%
Sample 1	2.6216	1.1612	3.9699	5.1877	1.8962	7.2879	+83.58
Sample 2	2.4696	1.3706	4.0410	7.3088	3.1841	11.0102	+172.46
Sample 3	2.9537	2.0805	5.3131	4.4474	2.1003	6.8768	+29.43
Sample 4	3.4790	1.6895	5.4031	9.1615	4.1023	13.922	+156.39
Sample 5	2.3951	1.5955	4.2088	5.2145	3.0457	8.6974	+66.79
Average			4.5926			9.5589	+66.79

(Chlorophyll Content –Mean of two replications in milligrams /gram/ fresh weight)
The above tests were conducted under the aegis of the Department of Biochemistry , College of Agriculture, Anand Campus , Gujarat Agricultural University , Gujarat India .

Table -4
EFFECT OF SOIL AMENDMENTS AND FOLIAR SPRAYS OF BIO MAGIC ON YIELD OF SUMMER GROUNDNUT

	TREATMENT	POD YIELD		POD	% INCREASE
		1992	1993		
1	BIO MAGIC (Soil)	1151	2745	1948	+5.01%
2	BIO MAGIC (Foliar)	1533	3139	2337	+25.96%
3	Ferrous Sulphate (Soil)	1007	2385	1696	-8.57%
4	Ferrous Sulphate +Ca (Foliar)	1226	2782	2004	+8.03%
5	25kg N/Ha	1408	2752	2080	+12.13%
6	Absolute Control	1082	2627	1855	0.00

Considering two years pooled results, foliar spray of BIO MAGIC was significantly superior to other treatments, whereas in individual years treatment differences were non- significant but when degree of freedom increased the difference came to be significant in pooled data. The data presented in this experiments is quite in agreement with the recommendation already adopted in joint Agresco (May 93) when 28.0% increase in yield was reported . Soil treatment with ferrous sulphate Of BIO MAGIC proved non effective.

Table 5
EFFECT ON GROWTH STIMULANTS ON THE YIELD OF GROUNDNUT

TREATMENT	POD YIELD(Kg/Ha)	% CHANGE OVER CONTROL
NAA @ 20ppm	2649	+6.0%
G.A.@ 20ppm	2367	-5.3 5
Boric Acid @ 0.2%	2642	+5.7%
AKTI M -3	3275	+31.0%
FeSO4+ Citric Acid (5:1)	2770	+10.2%
Control (No Spray)	2499	0.00
S. Em	162.6	
C.D. at 5%	489.7	
C.V %	12.03	

The above data indicate the superiority of BIO MAGIC -3 sprays at intervals of 15 days commencing from 30 DAS registering 31% increase in yield. This experiment also showed that conventional treatments of FeSo4+ Citric Acid spray was not as effective in increasing yield as BIO MAGIC , The data also occur with earlier recommendation adopted at Joint Agresco, May 1993, when a 28% increase in yield was recorded. Considering the results ,i.e. three years in Agronomy Department (from where the recommendation emerged), two years collecting data from experiments undertaken based on suggestion from committee on chlorosis in groundnut regulators, BIO MAGIC has proved itself an effective agent in increasing the yield of groundnut by between 25 to 30%.

RESULTS OF PLOT AND MINI- PLOT TRAILS ON GROUNDNUT WITH BIO MAGIC SPRAY (CY 2005-2006)

		PLOTS				MINI-PLOTS			
TRAIT OBSERVED	MATURITY	CONTROL	TREATED	% INCREASE	% DECREASE	CONTROL	TREATED	% INCREASE	% DECREASE
No of pods	Mature	4.9	7.9	53	-	12.5	20.8	66	-
	Immature	6.1	4.1	-	32	10.8	4.71	-	55
No of Kernels	Mature	7.18	9.0	25	-	16.8	35.2	109	--
	Immature	8.36	7.27	-	13	12.0	8.0	-	33
Wt of pods(g)	Mature	3.8	5.1	34	-	13.39	24.13	80	-
	Immature	2.6	0.8	-	69	2.93	1.32	-	54
Wt of Kernels	Mature	2.6	2.7	8	-	7.81	7.81	111	-
	Immature	1.9	1.5	-	21	2.02	2.02	-	29

Above figures are per plant

Conclusion: The results of the above replicated trials indicate that there is a significant increase in all the desirable parameters essential for increase in yield of groundnut. There is a net increase in number and weight of mature pods with decrease in number and weight immature pods with BIO MAGIC treatment. This could be due to supply of photosynthate required during all the stages of plant growth resulting in net increase yield – the sum total effect of increased chlorophyll in treated and plots.

RESULTS OF EXPERIMENTS CONDUCTED ON FARMERS' FIELDS AT 22 LOCATIONS IN JUNAGADH DISTRICT, GUJARAT , INDIA ,IN 2004

SHOWING EFFECT OF BIO MAGIC ON GROUNDNUT CONDUCTED UNDER THE AGA KHAN RURAL SUPPORT PROGRAMME

NO	NAME OF VILLAGE	NO.OF LOCATIONS	AVERAGE YIELD(kg/Acre) CONTROL	TREATED
1	Hussainabad	4	750	825
2	Samadhiyalaya	4	550	600
3	Hirenvel	4	500	550
4	Sukhpur	5	500	570
5	Chanduvav	5	500	575
			2800	3120
			560	624

Yield of dry pods of groundnut in Kg/acre

As per of programme organizer, the Aga Khan Rural Support Programme there was **50% recovery from CHLOROSIS after first spray and more than 90% after second and third sprays.** Despite erratic/ untimely rainfall during the growing season, there was a **net increase of 11.4% in the yield** of groundnut pods.

RESULTS OF EXPERIMENSTS CONDUCTED ON FARMER'S FIELDS OF BHAVNAGAR DISTRICT DURING THE SUMMER OF 1991 BY T & V STATE DEPARTMENT OF AGRICULTURE. - GUJARAT STATE

NO	FARMER'S NAME	VILLAGE	TALUKA	NO & DATE OF SPRAYS (AKTI M3) (1991)	YIELD IN PODS (KG/HA) TREATED	CONTROL	REMARKS OF DEPUTY DIRECTOR OF AGRICULTURE
1	Madhu D.	Tarad	Mahuwa	Three 29/3, 8/4, 29/4	2430	2165	Very Good results in Yield
2	Arvind. N.	Savarkundla	Savarkundla	One 30/3	1735	1725	No Advantage
3	Nanu V	Savarkundla	Savarkundla	One 31/3	1470	1470	No Advantage
4	Navnit. V	Kundheli	Talaja	Two 30/3, 9/4	2630	2300	Good Results
5	Shamu T	Kundheli	Talaja	Two 31/3, 10/4	2565	2142	Good Results
6	Devji L.	Kundheli	Talaja	Two/ 31/3, 10/4	2700	2435	Good Results
7	Premji P.	Kundheli	Talaja	Two/ 29/3, 8/4	1980	1640	Good Results
8	Damji V.	Kundheli	Talaja	Two/ 31/3, 9/4	2190	1910	Good Results
	AVERAGE				2212	1973	Percentage increase 12% *

- If we disregard the data of the farmers who have sprayed only once then the percentage increase in yield is 15%.

RESULTS OF EXPERIMENTS CONDUCTED ON FARMER'S FIELDS UNDER THE OILSEED PRODUCTION PROGRAMME IN KHARIF,1991.GROUNDNUT CROP GROWN IN PROBLEM AREAS OF RAJKOT DISTRICT, GUJRAT STATE TREATED WITH AKTI-M-3 , 0.5% SOLUTION BY FOLIAR APPLICATION

NO	TALUKA	NO OF	GROUNDNUT PODYIELD*	INCREASE IN

	(SUBDISTRICT)	TRAILS	TREATED	CONTROL	YIELD(%)
1	Kandorna	4	170	155	9.68
2	Dhoraji	4	580	515	12.62
3	Kothda Sangani	4	425	402	5.72
4	Upleta	4	510	400	27.50
5	Jetpur	4	660	645	2.33
6	Gondal	5	1170	1010	15.84
	Total	25			

*Yield inn Kg/Plot , The results are drawn from the report of Extension officer, T& V Jetpur division. The plot sizes varied. All the trails conducted in one Taluka were considered one unit and percentage increase or decrease in yield as a parameter for comparison. Despite scarcity of rain in Kandoena, Kota Sangani and Jetpur talukas, and considering that in some instances the crop was harvested earlier (due to failure of monsoon rains) the above data clearly indicate a net 12.30% increase in yield even at district level in 25 trails at diverse locations. At locations where protective irrigation was possible, the results are very encouraging.

THE STUDY OF COMPARATIVE PERFORMANCE OF DIFFERENT GROWTH STIMULANTS ON SUMMER GROUNDNUT

NO	TREATMENT			YIELD DRY PODS (KG/HA)				
	STIMULANT	DILUTION	APPLICATION	1990*	1991**	1992***	POLLED	SHELLING%
1	NAA	25 PPM	30 DAS	2933BC	2828AB	1742BC	2501CD	62.8
2	NAA	25 PPM	30 & 40 DAS	3048 AB	2684BC	2138 B	2624 BC	64.4
3	GA	25 PPM	30 DAS	3007ABC	2836AB	1820BC	2554C	64.1
4	GA	25 PPM	30 & 40 DAS	2970ABC	2821AB	1959BC	2584BD	64.4
5	Boric Acid	0.2%	30 DAS	3244 A	2960AB	2072B	2759B	67.7
6	Boric Acid	0.2%	30& 40 DAS	3089AB	2799AB	1542C	2476CD	62.7
7	Photosynthegin	20 PPM	30 DAS	3113AB	2836AB	1939BC	2629BC	61.1
8	Photosynthegin	20 PPM	30 & 40 DAS	2983ABC	2769AB	1725BC	2492CD	63.1
9	BIO MAGIC	0.5%	30,45&60 DAS	3175AB	3031A	2784A	2996A	67.1
10	Control			2765C	2435C	1808BC	2336D	65.5
	S.E.+/-			87.68	92	144	60.6	
	C.D.+/-			248	260	406	194.0	
				10	11.4	22	5	

Data courtesy of Mr.J.J.Macwan, Ph.D. Thesis (Agronomy) ** Data from Agronomy Dept , Gujarat Agricultural University , Anand, India. Interpretation consistent performance of BIO MAGIC merits recommendation for farmers,

APPENDIX IV A : RESULTS OF TRAILS CONDUCTED ON GROUNDNUT CROPS AT VERAVAL IN 1991 UNDER THE AGA KHAN RURAL SUPPORT ROGRAMME (TALUKA MANGROI, DISTRICT :JUNDAGADH)

NO	NAME Of FARMER	VILLAGE	YIELD(Kg/Acre) CONTROL	TREATED	INCREASE IN YIELD(%)
1	Bhana Rama	Karamadi	1000	1135	13.5
2	Bikha Devanand	Karamadi	760	1000	31.5
3	Rama Naran	Kankasa	Vitiated*		
4	Bhura Ramanlal	Kankasa	No harvest at all due to drought		
5	Naran Natha	Kankasa	135	165	22.2

Experiment was vitiated as farmer applied concentrated ferrous sulphate solution, one week after application of BIO MAGIC spray . Average increase in yield was 21.5%

APPENDIX IV B , RESULTS OF TRAILS CONDUCTED ON GROUNDNUT CROPS AT VERAVAL, IN 1992 UNDER THE AGA KHAN RURAL SUPPORT PROGRAMME (TALUKA: MANGROI, DISTRICT: JUNAGADH)

NO	NAME OF FARMER	VILLAGE	YIELD (KG/ACRE) CONTROL	TREATED	INCREASE IN YIELD(%)
1	Bhura Ranmal	Karmadi	480	600	25.0
2	Raju Karsan	Karmadi	600	720	20.0
3	Kana Punja	Karmadi	600	690	15.0

Out of six, three experiments were vitiated due to continuous water logging.Average increase in yield was 20.0%.

RESULTS OF EXPERIMENTS CONDUCTED ON FARMER'S FIELD'S IN KHEDA DISTRICT, GUJARAT , INDIA, 1992. FOLIAR SPRAY OF BIO MAGIC ON SUMMER GROUNDNUT (AS PER RECOMMENDATION OF STATE LEVEL COMMITTEE)

NO	TALUKA	NO.OF TRAILS	GROUNDNUT PODYIELD* TREATED	CONTROL	INCRESE IN YIELD (%)
1	Balasinor	3	1623	1303	24.6
2	Khambhat	6	2162	1996	08.3
3	Anand	4	3575	2865	24.8
4	Thasra	1	2160	2400	-11.1
	Weighted mean	14	2336	2031	15.0

*Groundnut pod yield in kg/Ha ; Conclusion: Results of BIO MAGIC spray on summer groundnut in Kheda district indicate overall 15% increase in yield even after considering negative result obtained at one place because of poor management (about one half of the treated crop was eaten by cattle!) as per report from the Training & Visiting (Agricultural) officer.

APPENDIX VI : COMPARATIVE PERFORMANCE OF BIO MAGIC WITH CONVENTIONAL TREATMENT OF FERROUS SULPHATE+ CITRIC ACID AT GUJARAT AGRICULTURAL UNIVERSITY (ANAND CAMPUS), INDIA. CROP: SUMMER GROUNDNUT

NO	TREATMENT	DRY POD YIELD (Kg/Ha)	SHELLING (%)
1	BIO MAGIC Soil Application at 5 liters/Ha	1239	63.3
2	BIO MAGIC Foliar spray (3 sprays at 30,45 & 60 DAS)	1580	66.2
3	Ferrous Sulphate Soil Application at 50 kg/Ha	1110	63.5
4	Ferrous Sulphate + Citric Acid at 10+2 Kg/Ha (3 applications at 30,45 & 60 DAS)	1077	63.6
5	Nitrogen at 25 Kg/Ha	1361	66.1
6	Absolute Control	1041	64.0
	S.E.m		
	C.D.5%		
	C.V.%		

CONCLUSION: Foliar spray of BIO MAGIC recorded highest pod yield and was significantly superior to application of ferrous sulphate + citric acid. Soil application of BIO MAGIC and ferrous sulphate did not produce significant results.

APPENDIX VII
June 29th , 1996.

WALLACE LABS
TISSUE REPORT
165 Coral Circle,
EI Segundo, CA 90245
(310) 615-0116

TISSUE REPORT
Location Portland, OR
Requester: Kurt Anderson

ELEMENTS	**96-180T-1 shoots**	**96-180T-2 179T-Roots**	**Average Lettuce**
Phosphorus	7,472.47	13,019.88	8,000
Potassium	65,212.08	71,584.79	80,000
Iron	148.80	271.02	75
Mamganese	226.03	106.55	50
Zinc	152.72	136.08	40
Copper	13.36	14.52	10
Boron	36.77	19.72	40
Calcium	16,627.89	8,940.40	30,000
Megnesium	2,553.56	1,955.58	7,000
Sodium	683.15	1,074.42	
Sulfar	2,910.20	2,066.66	
Molybdenum	0.12	0.21	
Aluminium	109.25	253.05	
Arsenic	< 0.13	0.33	
Barium	53.97	69.03	
Cadmium	1.47	0.84	
Chromium	0.40	0.50	
Cobalt	<0.05	<0.05	
Lead	1.81	5.21	
Lithium	1.77	14.30	
Mercury	<0.03	<0.03	
Nickel	4.50	4.18	
Selenium	0.49	0.49	
Silicon	120.34	99.78	
Silver	<0.02	<0.02	
Strontium,	87.88	81.56	
Tin	<0.28	<0.28	
Titanium	4.31	9.79	
Vanadium	0.29	0.81	
Nitrogen			
Chloride			

Units are mg/kg (parts per million) on a dry weight basis expect as noted.
Leaves and roots were rinsed with deionised water

APPENDIX VII WALLACE LABS, 165 Coral Cirlce,El Segundo, CA 90245

ELEMENTS	96-14T-1	96-14T-2	RADISH – STANDARD VALUES			UPTAKE BY SHOOTS (in mg)		
	TREATED	UNTREATED	LOW	SUFFICIENT	HIGH	TREATED Shoots	UNTREATED Shoots	% INCREASE (by treatmt)
Phosphorus	3,174.0	3682.32	2,500-2,900	3,000-7,000	>7,000	3.6627	0.8703	421%
Potassium	38,922.2	28397.93	35,000 39,900	40,000-75,000	>75,000	36.7918	6.7114	548%
Iron	598.2	514.07	40-49	50-200	>200	0.5669	0.1279	443%
Manganese	44.5	38.80	35-49	50-100	>100	0.0422	0.3092	460%
Zinc	227.1	174.08	15-18	19-100	>100	0.2150	0.0411	522%
Copper	16.0	12.37	3-4	5-25	>25	0.0155	0.0029	530%
Boron	25.8	26.04	20-24	25-125	>125	0.0245	0.0062	398%
Calcium	22,589.0	21,473.74	20,000 29,900	30,000-45,000	>45,000	21.3523	5.0750	421%
Magnesium	3,742.7	3,348.01	3,000-4,900	5,000-12,000	>12,000	3.5372	0.9912	447%
Sodium	1,462.6	736.79				1.3829	0.1741	794%
Sulfur	5,999.4	4,650.68				5.6708	1.0991	516%
Molybdenum	4.7	7.18				0.0038	0.0017	227%
Aluminium	1,058.0	943.16				1.0005	0.1229	449%
Arsenic	0.6	0.83				0.0008	0.0002	413%
Barium	59.9	73.19				0.0561	0.0173	325%
Cadmium	0.4	0.15				0.0004	0.0000	1186%
Choromium	1.4	1.13				0.0012	0.0003	440%
Cobalt	<0.5	<0.05						
Lead	7.7	6.69				0.0069	0.0016	435%
Lithium	1.3	1.12				0.0012	0.0003	441%
Mercury	<0.3	<0.03						
Nickel	2.6	1.43				0.0020	0.0003	578%
Selenium	<0.9	0.49						
Silicon	391.0	374.23				0.3699	0.0884	418%
Silver	0.3	<0.02				0.0002		
Strontium	124.8	135.98				0.1178	0.0321	366%
Tin	0.8	<0.28						
Titanium	20.5	20.75				0.0196	0.0049	400%
Vanadium	2.1	2.00				0.0021	0.0005	444%
Nitrogen	4.9%	6.45%	2.8%-3.0%	3.0%-6.0%	>6.0%	46.7899	15.2435	307%

Units are mg/kg (parts per million) on a dry weight basis expect as noted.
Requester: Kurt Anderson Dry weight per plant in grams. Plant: Radish

Shoots	0.95	0.236
Roots	0	0.17

EFFECTVENSESS OF METHODS DEVELOPED AND EMPLOYED FOR CONTROLLING CHLOROSIS BY USE OF BIO MAGIC.

INTRODUCTION

In chlorosis affected leaves thylacoids- organelle where chloroplasts manufactured – are badly affected and do not produce adequate chlorophyll. Grana sacks are greatly reduced (from the desired 12 to 14 to a mere 3 to 7) rendering chloroplasts inefficient in utilizing solar energy thereby affecting the process of photosynthesis. For managing such situations frequent spraying of micronutrients –containing formulations are widely used but this practice has certain demerits. Another approach is to apply soil amendments, mainly use of sulphur containing fertilizer, copious supply of FYM, acidification of soil etc. Despite this the desired results are not always obtained. In advanced countries pressure injections in tree trunks are practiced but these are expensive, require skilled technicians and hence are cost – prohibitive. We therefore tired to develop methods which are simple, easy to practice and cost effective. And after prolonged experimentation, research –both in the laboratory and in the field , continuous monitoring , assessment and review , we are confident that in BIO MAGIC we have a product that addresses the problem, shows results within 72 hours, is simple to use and is cost- effective. In addition, being of origin, it is non –poisonous and eco – friendly. The following methods of application of BIO MAGIC were all tired.

SPOT APPLICATION

This method id primarily meant for studying effectiveness in absorption, penetration, proliferation and time required for regreening of treated parts. Twings of chloratic trees are selected keeping in view age and degree of chlorosis of trees to be treated. In bifurcated twigs one branch is kept as treated and another as control and continuous monitoring is done to study regreening. In the branch selected for treatment a list is made in the bark and BIO MAGIC was applied with specific carrier material.

DRIP DROP FEEDING

In this method a hole is drilled (depending on diameter of branch to be treated) and BIO MAGIC liquid (diluted 5 times) is dropped into the hole at three drops per minute.(IV sets discarded by hospitals /clinics were used for this purpose as they worked out cheapest.)

ROOT FEEDING

This is useful in shallow rooted trees like citrus , coconut ,fig , cashew nut , grapes, etc. Absorbing root (about 1 cm in diameter) is selected and a cut with a desired angle is made on the underside. BIO MAGIC (duluted 5 times) is fed to absorbing root. This last method ,viz Root Feeding is still under trail and various experiments are in progress. Results communicated to us are highly encouraging ,but photographic records are not available yet. Reports from Nangadon in Maharashtra , India indicate hat coconut trees treated by root feeding have flowered for the first time in ten years. As soon as research and trails completed, photographic evidence to demonstrate the effectiveness of this method will be made available.

EFFICIENCY OF BIO MAGIC- TRIAL REPORTS AND RESPONSES FROM SPECIFIC CROPS TO BIO MAGIC TREATMENT

EFFICIENCY TESTING OF BIO MAGIC ON SAPOTA (CHICU)

Sapota has been reported to suffer from chlorosis. Shorrocks (1984) has included Sapota in the list of trees sensitive to chlorolsis. Trees affected by chlorosis bear fruit of smaller size and / or in lesser number and although leaves may not show the yellow coloring they are of a paler green compared to the leaves of normal trees.

EFFICIENCY TESTING OF BIO MAGIC ON MANGO (MANGIFERA INDICA)

INTRODUCTION:

Mango (Mangifera Indica) has been reported to be affected with chlorosis especially in orchards raised on carcareous soils. Mango is also a tropical evergreen tree and 75% of fruit are produced in India. Chlorosis is often associated with die-back disease having unknown etiology. Recently :Bandi" (local name of disease in "Kesar" growing area of Gujarat state , India) is reported on the increase. In this disease young leaves at the terminal portion of branches become chloratic and malformed and sometimes fall off giving cactus – like appearance to terminal branches. The tip of the effected branch becomes necrotic and eventually dies. New buds do sprout from lower side of affected branch but usually meet the same fate. The Kesar variety of mango has a delicious taste and pulp of golden colour. It also has specifically appealing flavour, sweetness and keeping quality and has earned credit and is great demand from both Indian and export markets. A survey of the area showed that more than 50% of the trees were affected with " Bandi". Not all trees in every orchard were affected. Trees not affected were healthy and showed normal yield performance. "Bandi" also strikes in young orchards and sometimes all the saplings in an orchard have to be uprooted and new ones planted- causing a loss of time and money.

At village Rampol, about 25km from Anand, a farmer who had an orchard of 216 Kesar mango trees decided to destroy the orchard because all the saplings (about 3 to 4 years old) showed die – back symptoms. We requested him not to uproot the saplings started to sprout new buds with dark green leaves and subsequently all the saplings were treated with SA. Now he has decided not to uproot the saplings.
Similarity, at village Talala of Junagadh District which is the heartland of Kesar mango, "Bandi" was observed at several orchards. Following the successful treatment at Ramol we also undertook to treat the trees at Talala. Later we had reports of "Bandi" in other areas in Gujarat and all were treated on a trail basis (SA, DD and RF). Results and feedback coming from various locations are encouraging. There are indications that such kind of treatment may prove helpful in chlorosis induced die- back and restoring orchards.